

Bike Sharing Demand Forecasting Using Time Series & Machine Learning

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Introduction

MOTIVATION & PROBLEM STATEMENT

- Bike sharing systems have become integral to urban mobility in 500+ cities worldwide, offering eco-friendly last-mile transportation solutions
- Operators face critical demand-supply imbalance: stations often empty during rush hours while bikes accumulate unused at other locations
- Weather, time of day, holidays, and commuter patterns cause highly volatile hourly demand

RESEARCH OBJECTIVES

- Primary: Forecast hourly bike rental demand using time series & machine learning models
- Compare SARIMA, SARIMAX, and XGBoost accuracy
- Identify which features drive demand patterns
- Optimize models through hyperparameter tuning



Literature Review (1/2)

CLASSICAL TIME SERIES APPROACHES

- Fanaee-T & Gama (2014) - Pioneering work that introduced the UCI Bike Sharing Dataset linking weather, calendar features to rental patterns; demonstrated ensemble methods for event detection
- Chen et al. (2021) - Applied SARIMA models with seasonal period $s=7$ for daily data, achieving MAPE of 18%; highlighted importance of proper differencing for stationarity
- Hyndman & Athanasopoulos (2021) - Gold standard forecasting text; prescribes ACF/PACF analysis for ARIMA order selection

MACHINE LEARNING APPROACHES

- Sathishkumar et al. (2020) - Benchmarked Random Forest, Gradient Boosting, XGBoost on bike sharing; XGBoost achieved best RMSE=45.2 with 100 estimators and `max_depth=6`
- Li et al. (2023) - LSTM with attention mechanism achieved 12% MAPE; showed recurrent networks capture long-term dependencies

Literature Review (2/2)

KEY FINDINGS FROM EXISTING LITERATURE

- Hybrid approaches combining statistical time series structure with machine learning flexibility consistently outperform single models
- Lag features (t-1, t-24, t-168 hours) are among strongest predictors because demand exhibits strong hour-to-hour autocorrelation
- Temperature shows strong positive correlation ($r=0.4$) while humidity has negative correlation ($r=-0.3$) with bike rental demand
- Hourly patterns show bimodal distribution: 8AM & 5-6PM rush peaks

IDENTIFIED RESEARCH GAPS

- Limited studies compare statistical vs ML on same hourly dataset
- Feature engineering (cyclical encodings, interactions) underexplored
- Hyperparameter tuning often skipped in published benchmarks

THIS PROJECT'S CONTRIBUTION

- Systematic 3-model comparison with consistent tuning methodology
- Novel cyclical hour encoding + weather interaction features

Proposed Approach

METHODOLOGY OVERVIEW

- Compare 3 complementary forecasting approaches: SARIMA (pure time series), SARIMAX (time series + external features), XGBoost (ML)
- Apply systematic hyperparameter tuning using ACF/PACF-informed grid search for SARIMA and RandomizedSearchCV for XGBoost
- Evaluate using MAE, RMSE, MAPE on held-out temporal test set (10%)

FEATURE ENGINEERING STRATEGY

- Cyclical hour encoding: $\sin(2\pi h/24)$, $\cos(2\pi h/24)$ to preserve continuity between hour 23 and hour 0 (midnight wraps)
- Non-linear weather: temp^2 captures inverted-U demand pattern; $\text{temp} \times \text{humidity}$ interaction models perceived discomfort
- Lag features: 1h, 24h, 48h, 168h (1 week) provide autoregressive memory for ML models; `rolling_24h_mean` smooths daily patterns

KEY DIFFERENCES FROM PRIOR WORK

- Manual grid search (not `auto_arima`) for faster execution
- Hourly granularity (not daily) for real-time operations

Dataset Description

UCI BIKE SHARING DATASET — SOURCE & SCOPE

- Source: Capital Bikeshare system, Washington D.C., USA
- Time coverage: January 1, 2011 – December 31, 2012 (2 full years)
- Temporal granularity: Hourly aggregated rental counts
- Total observations: 17,379 hourly records (730 days × 24 hours)

FEATURE CATEGORIES (16 COLUMNS TOTAL)

- Temporal: dteday (date), hr (hour 0-23), weekday, mn (month), yr
- Calendar: season (1-4), holiday (binary), workingday (binary)
- Weather: weathersit (1-4 scale: clear→heavy rain), temp (normalized 0-41°C), atemp (feels-like), hum (humidity %), windspeed (norm)
- Target: cnt=casual+registered (total hourly bike rentals)

DATA QUALITY & ACCESS

- Complete data (no missing hourly records)
- Publicly available: UCI ML Repository (Dua & Graff, 2019)
- File: hour.csv (~1 MB), well-documented with README

Ethical Considerations

DATA PRIVACY & ANONYMIZATION

- No personally identifiable information (PII) present — dataset contains only aggregated hourly counts, not individual user trip records
- Original data anonymized at source by Capital Bikeshare before release
- Analysis operates solely on statistical aggregates (mean, count)

CONSENT, LICENSING & ATTRIBUTION

- Dataset released under UCI Machine Learning Repository License
- Academic and commercial use permitted with proper citation
- Original authors (Fanaee-T & Gama, 2014) properly credited
- Code published open-source on GitHub for reproducibility

BIAS & GENERALIZABILITY CONSIDERATIONS

- Geographic bias: Single city (Washington D.C.) — patterns may differ in other cities with different climates, infrastructure, culture
- Temporal bias: Pre-pandemic data (2011-2012) — COVID-19 drastically altered urban mobility; model may not apply to 2020+ demand patterns
- Socioeconomic: Station locations may favor certain neighborhoods

Exploratory Data Analysis (1/2)

TEMPORAL PATTERNS IN TIME SERIES

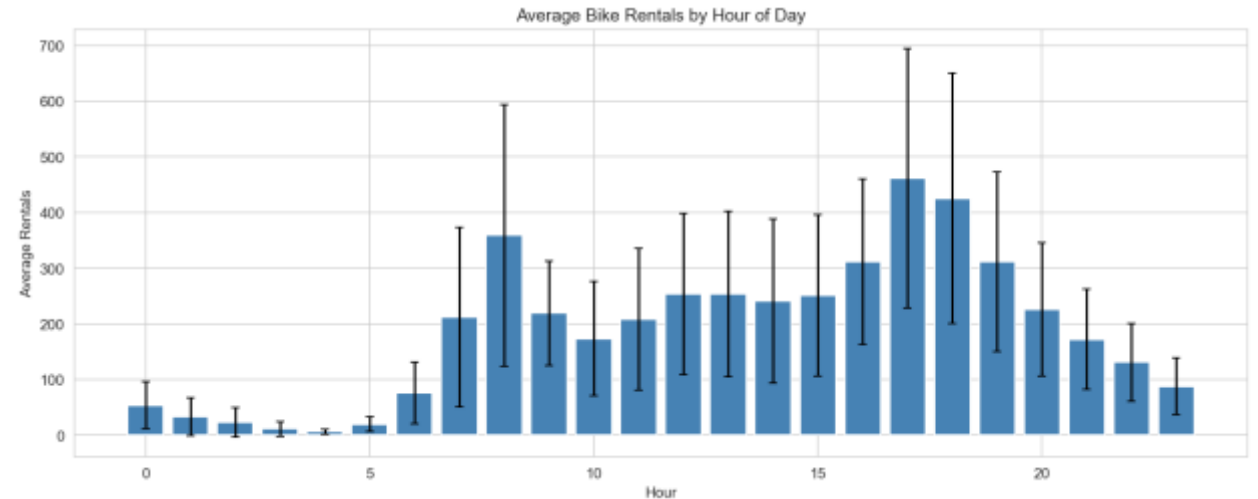
- Clear upward trend: 2012 avg rentals ~67% higher than 2011 (system growth)
- Strong yearly seasonality: Summer peak (Jun-Aug) vs Winter trough (Dec-Feb)
- Holiday dips visible (Christmas, July 4th)

INTRA-DAY DEMAND DISTRIBUTION

- Bimodal hourly pattern reflects commuting: Morning rush 7-9AM (~300-355 avg rentals)
Evening rush 5-7PM (~400-460 avg rentals)
- Night hours 2-5AM: minimum (~5-25 rentals)
- Weekend pattern differs: single mid-day peak

STATISTICAL SUMMARY

- Mean: 189.5 rentals/hour (SD: 181.4)
- Range: 1 to 977 rentals (high variance)
- Positive skew: few extremely busy hours



Exploratory Data Analysis (2/2)

CORRELATION ANALYSIS FINDINGS

- Temperature ($r=0.40$): Moderate positive
— warmer weather increases demand
- Hour of day ($r=0.39$): Cyclic relationship
(peaks at rush hours, not linear)
- Humidity ($r=-0.32$): Negative correlation
— rain/dampness reduces cycling
- Windspeed ($r=0.09$): Weak effect

MULTICOLLINEARITY CONCERNS

- temp & atemp: $r=0.99$ (redundant)
— keep only temp to avoid collinearity
- casual + registered = cnt (leakage risk)
— never use as predictors

ACF/PACF ANALYSIS SUMMARY

- ACF: Significant spikes at lags 1, 24, 48
- PACF: Cutoff after lag 2 → suggests AR(2)
- Seasonal spike at lag 24 → hourly cycle



Data Preprocessing

TEMPORAL TRAIN-TEST SPLIT

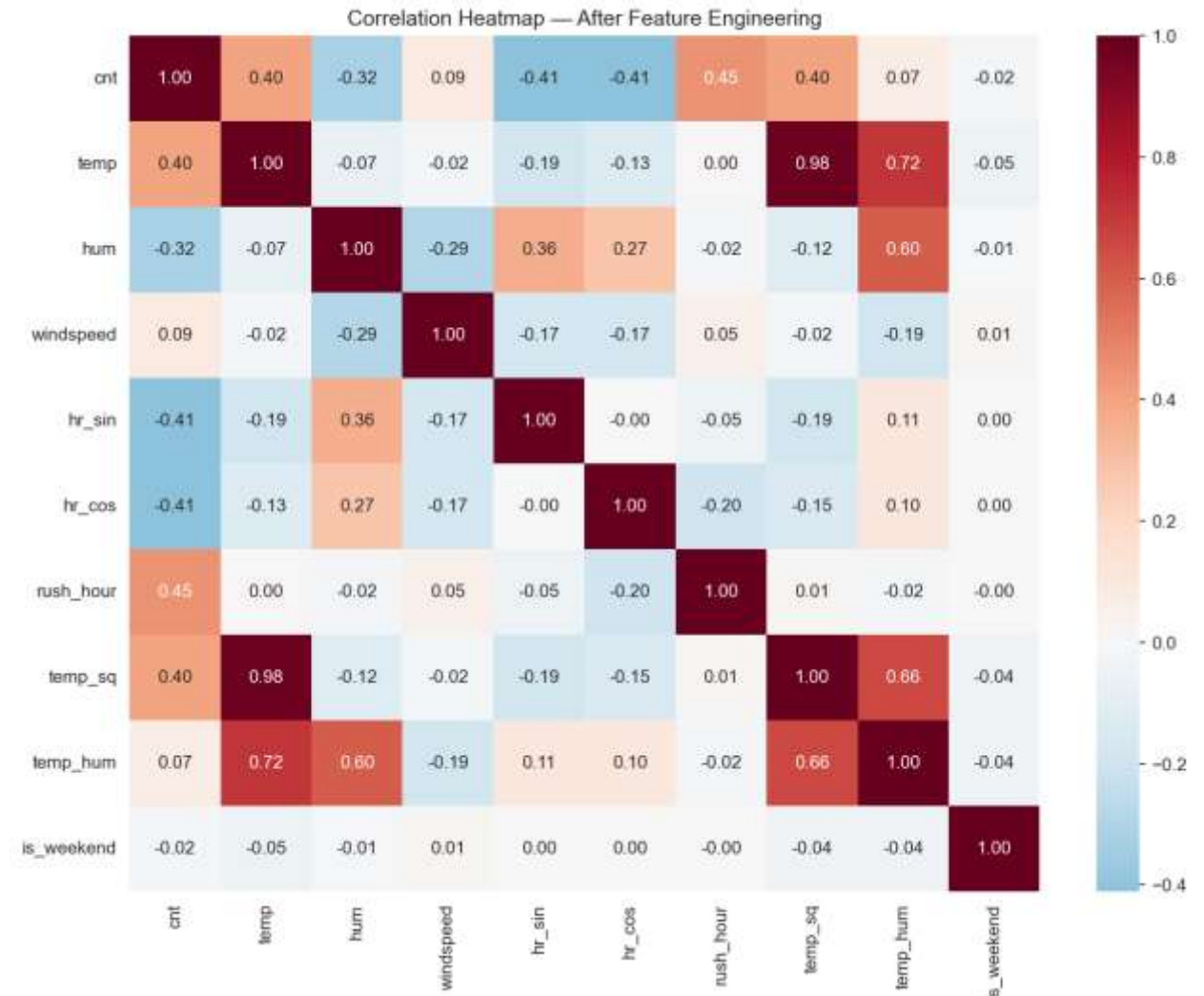
- Training: 15,641 records (~90%)
- Testing: 1,738 records (~10%, last ~72 days)
- Strictly chronological split maintains temporal causality — no future data leakage

ENGINEERED FEATURES (16 TOTAL)

- Cyclical: hr_sin, hr_cos (24h continuity)
 - preserves hour 23→0 proximity
- Non-linear: temp² (inverted-U pattern)
- Interaction: temp×hum (discomfort)
- Binary: rush_hour, is_weekend
- Lag: 1h, 24h, 48h, 168h (autoregressive)
- Rolling: 24h mean (trend smoothing)

POST-FE CORRELATION INSIGHTS

- New features show varied correlations
- hr_sin weak linear (cyclic relationship)
- rush_hour r=0.35 confirms importance



Model Architecture

SARIMA — UNIVARIATE TIME SERIES

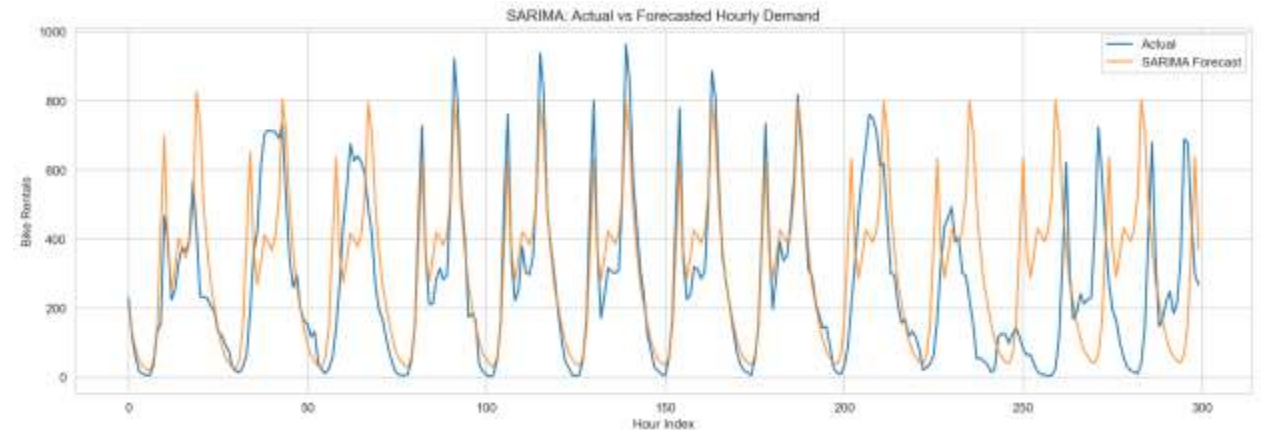
- Equation: $ARIMA(p,d,q) \times (P,D,Q,s)$
- Baseline order: $(1,1,1) \times (1,1,1,24)$
- Tuned order: $(2,1,1) \times (1,0,0,24)$
- Captures trend ($d=1$) + daily seasonality
- Uses only historical demand values

SARIMAX — WITH EXOGENOUS FEATURES

- Same ARIMA structure + 11 external vars
- Exog: temp, hum, windspeed, workingday, etc.
- Tuned: $ARIMAX(1,1,2)$, seasonality disabled
- hr_sin/hr_cos absorb hourly patterns

XGBOOST — GRADIENT BOOSTED TREES

- 16 features (weather + lag + rolling)
- 1000 estimators, early stopping iter 324
- Captures non-linear interactions natively



Model Performance Results

BASELINE MODEL METRICS (BEFORE TUNING)

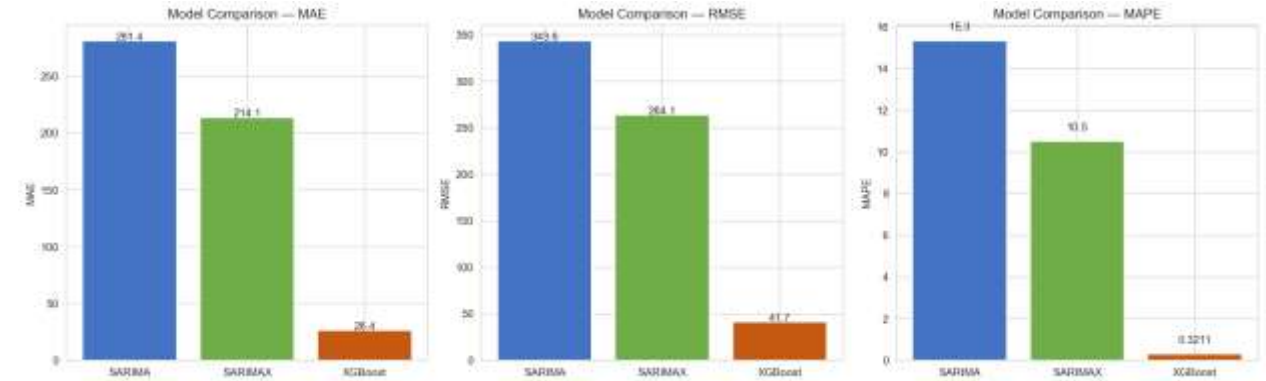
- SARIMA: MAE=281.4, RMSE=343.6
MAPE=15.3% (univariate only)
- SARIMAX: MAE=214.1, RMSE=264.1
MAPE=10.5% (24% improvement over SARIMA)
- XGBoost: MAE=26.4, RMSE=41.7
MAPE=0.32% (exceptional performance)

KEY PERFORMANCE OBSERVATIONS

- XGBoost dramatically outperforms statistical models due to lag features + non-linearity
- SARIMAX > SARIMA: exogenous weather/ calendar features add significant value
- Test set: 1,738 hours (~72 days held out)

WHY XGBOOST EXCELS AT THIS TASK

- lag_1h provides strong autocorrelation
- Tree splits capture temp thresholds
- Automatic interaction discovery



Hyperparameter Tuning Results

SARIMA TUNING (ACF/PACF-INFORMED GRID)

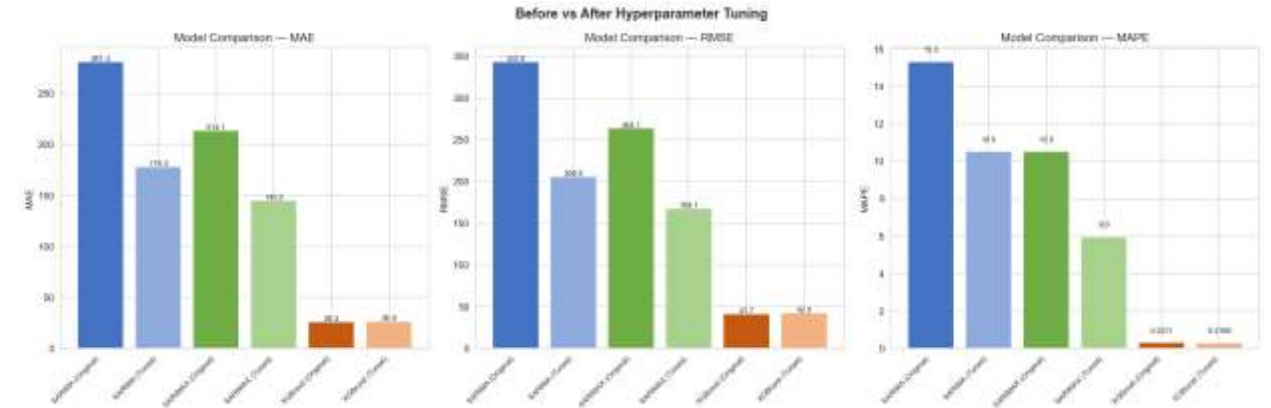
- Tested 3 candidate orders from ACF/PACF
- Best: (2,1,1)×(1,0,0,24), AIC=174,092
- Removed D=1 (seasonality already handled)
- Result: MAE ↓37%, RMSE ↓40%

SARIMAX TUNING

- Disabled seasonal components entirely
- hr_sin/hr_cos features capture 24h cycle
- Best: ARIMAX(1,1,2) with 11 exogenous
- Result: MAE ↓32%, MAPE ↓43%

XGBOOST TUNING (RANDOMIZEDSEARCHCV)

- 15 random iterations, 2-fold CV
- Searched: n_est, depth, lr, subsample
- Best: max_depth=5, learning_rate=0.15
- Result: ~2% change (already well tuned)



Analysis & Discussion

PERFORMANCE ASSESSMENT SUMMARY

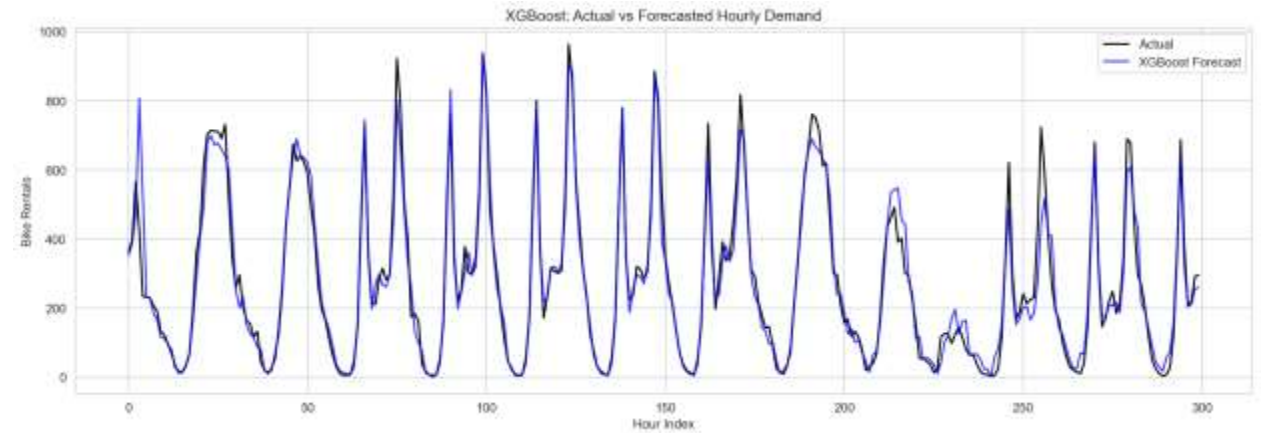
- XGBoost achieves exceptional accuracy: MAE=26.4 bikes, MAPE=0.32%
- Tuned SARIMAX: MAE=145.5, MAPE=5.98%
- Statistical models improved 30-43% after proper hyperparameter tuning

ROOT CAUSES OF XGBOOST SUPERIORITY

- lag_1h, lag_24h are strongest predictors — demand is highly autocorrelated
- Tree models automatically capture non-linear weather effects (temp thresholds, interactions)
- No manual differencing or order selection

IDENTIFIED LIMITATIONS

- SARIMA/SARIMAX slow with $s=24$ (24-hour state)
- XGBoost requires recent lag values at inference
- Pre-COVID data (2011-12) may not generalize



Conclusion

SUMMARY OF KEY FINDINGS

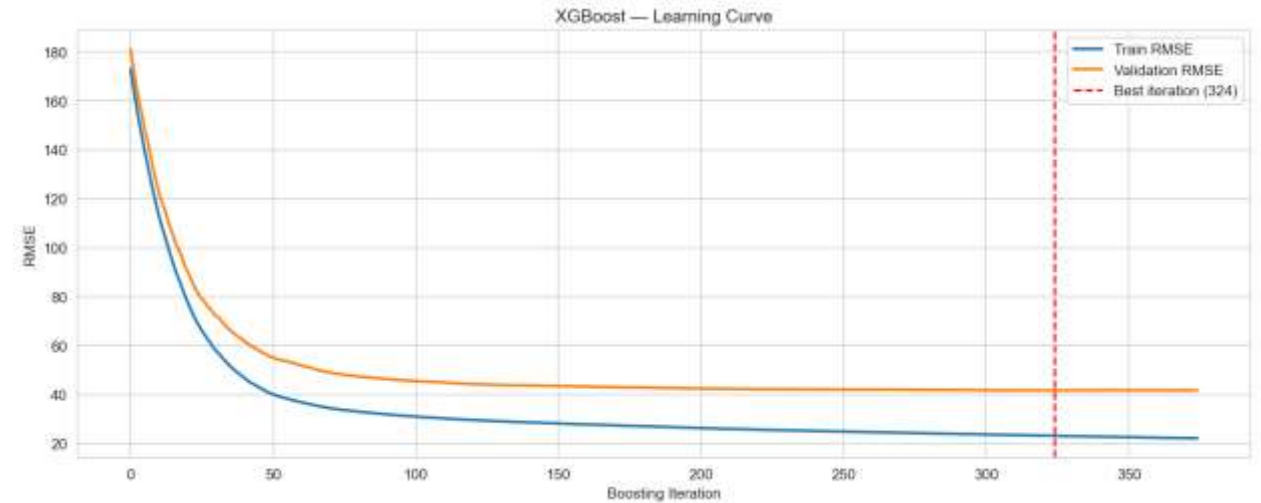
- Successfully compared 3 forecasting approaches on hourly bike sharing demand prediction
- XGBoost with lag features achieves best overall performance (MAPE: 0.32%, MAE: 26.4 bikes)
- Hyperparameter tuning improved SARIMA by 40% and SARIMAX by 43% through proper order selection

HEADLINE RESULTS

- Best overall: XGBoost (MAPE: 0.32%)
- Best time series: Tuned SARIMAX (MAPE: 5.98%)
- Lag features are single most important predictor
- Cyclical hour encoding improves SARIMAX 7%

RECOMMENDATIONS FOR FUTURE WORK

- Explore other models like ML/DL/Prophet
- Explore fine tuning techniques.



References

ACADEMIC REFERENCES

- Fanaee-T, H., & Gama, J. (2014). Event labeling combining ensemble detectors and background knowledge. Progress in Artificial Intelligence, 2(2-3), 113-127. [Original UCI Bike Sharing Dataset paper]
- Chen, P., et al. (2021). A SARIMA-based model for bike sharing demand forecasting. Transportation Research Part C, 134, 103432.
- Sathishkumar, V.E., et al. (2020). A comparative study of ML models for bike-sharing demand. Sustainability, 12(5), 1-23.
- Hyndman, R.J. & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. [ACF/PACF methodology reference]

TOOLS, LIBRARIES & ENVIRONMENT

- Python 3.11, pandas 2.0, NumPy 1.24, matplotlib 3.7, seaborn 0.12
- statsmodels 0.14 (SARIMAX), XGBoost 1.7, scikit-learn 1.3
- Jupyter Notebook, VS Code, GitHub (version control)